AEGISPHARM ENTERPRISE — TECHNOLOGY STACK

Complete Architectural Blueprint: Desktop Client, Web Server, Clinical CDSS & SQLite Engine

1. FRONTEND & DESKTOP

- **HTML5, CSS3 & ES6+ JavaScript**
  Modern reactive UI, CSS Variables
- **Microsoft Edge Chromium (WebView2)**
  Butter-smooth 60fps rendering
- **PyWebView 5.x Native Bridge**
  Zero-latency bi-directional RPC
- **Dark & Light Mode Engine**
  Local storage state persistence
- **Responsive Grid & Flexbox**
  Tailored for POS, Kiosks & Desktops
- **CustomTkinter 5.x**
  Native Python GUI fallback option

2. WEB APP & API SERVER

- **FastAPI 0.138.0**
  Asynchronous high-speed ASGI framework
- **Uvicorn 0.49.0 Server**
  Production-grade ASGI web server
- **REST & Universal RPC Hybrid**
  Universal /api/{method} dispatcher
- **OpenAPI & Swagger UI (/docs)**
  Interactive API explorer & ReDoc
- **Session Token Management**
  Independent multi-client sessions
- **Cross-Device Browser Support**
  Works on Chrome, Edge, Safari, Mobile

3. BACKEND & LOGIC LAYER

- **Python 3.12 Core**
  Robust, modern enterprise runtime
- **PharmacyBridge Controller**
  Unified business logic RPC wrapper
- **AppConnector Engine**
  Terminal presence heartbeats & telemetry
- **SettingsManager Singleton**
  Thread-safe dataclass config caching
- **Activity & Audit Logger**
  Centralized security audit streaming
- **Modular 6-Layer Architecture**
  Strict separation of concerns

4. CLINICAL CDSS & DOMAIN

- **CDSS Allergy Shield Engine**
  Penicillin, NSAID, Sulfa cross-match
- **Drug-Drug Interaction Matrix**
  Molecule contraindications analyzer
- **Schedule X NDPS Register**
  Dual-pharmacist PIN & doctor sign-off
- **FEFO Inventory Algorithm**
  First-Expiry First-Out dynamic batches
- **Cold-Chain IoT Simulation**
  2.0-8.0 C telemetry & thermal alarms
- **Multi-Tender POS & Taxes**
  Cash, UPI QR, Card & CGST/SGST calc

5. DATABASE & PERSISTENCE

- **SQLite 3 Embedded Database**
  Zero-config ACID-compliant SQL engine
- **10+ Relational SQL Tables**
  Staff, meds, rx, invoices, fleet, logs
- **Automated Seed & Schema Init**
  Self-bootstrapping database tables
- **JSON Configuration Storage**
  config.json with store & tax params
- **Database Snapshots Engine**
  Timestamped .db binary backups
- **CSV Audit Stream Exporter**
  Compliance audit trail reporting

6. SECURITY, QA & DEVOPS

- **Strict RBAC Access Control**
  5 Personas: Admin, Pharma, POS, Inv, Rider
- **Station Identity Lock**
  Prevents unauthorized role switching
- **Automated Test Suite (23 Tests)**
  100% pass rate across core & web
- **FastAPI TestClient & HTTPX**
  Integration testing for web & APIs
- **Git Version Control**
  Synchronized repository on GitHub main
- **Zero Cloud Dependencies**
  Fully functional on-premise & offline

AegisPharm Enterprise v2.5 • Dual Desktop & Web Architecture • 100% Python & Web Standards